

A Dynamic Traffic Signal Controlling System in Vehicular Networks

Md Shahriar Hossein and Sanjana Afrin Saima

A Thesis in the Partial Fulfillment of the Requirements
for the Award of Bachelor of Computer Science and Engineering (BCSE)



Department of Computer Science and Engineering
College of Engineering and Technology
IUBAT—International University of Business Agriculture and Technology

Summer 2025

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Letter of Transmittal

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Subject: Letter of Transmittal.

Dear Sir,

We are pleased to submit our thesis report titled “A Dynamic Traffic Signal Controlling System in Vehicular Networks”. This report has been prepared in partial fulfillment of the requirements for the degree of Bachelor of Computer Science and Engineering (BCSE) at IUBAT.

We sincerely hope that the report meets the academic standards and fulfills the requirements set by the department. We would be grateful for your kind consideration and evaluation.

Yours Sincerely,

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Student's Declaration

The thesis titled “A Dynamic Traffic Signal Controlling System in Vehicular Networks” is done for partial fulfillment of the requirements for the Award of Bachelor of Computer Science and Engineering (BCSE). It is the result of our research work under the supervision of Dr. Abhijit Saha, Professor, Department of Computer Science and Engineering, International University of Business Agriculture and Technology.

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Supervisor's Certification

This is to certify that the thesis entitled “A Dynamic Traffic Signal Controlling System in Vehicular Networks” submitted by Md Shahriar Hossein (21103082) and Sanjana Afrin Saima (22103101), has been carried out under my supervision. I declare this to be satisfactory for the partial fulfillment of the requirements for the award of Bachelor of Computer Science and Engineering (BCSE). They are now allowed to submit this report. I wish them the best of luck for their future undertakings.

Dr. Abhijit Saha

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Abstract

Vehicular Ad Hoc Networks (VANETs) enable communication between vehicles and roadside infrastructure, providing opportunities for data-driven Intelligent Transportation Systems (ITS). Conventional fixed-time traffic signals use predetermined signal timings regardless of the current traffic demand, which can lead to unnecessary waiting and inefficient use of available green time. This study presents a VANET-based Priority-based Dynamic Traffic Signal Control system that uses real-time vehicle information to adapt signal priority and green-time duration according to traffic conditions.

The proposed system incorporates On-Board Units (OBUs), Roadside Units (RSUs), and Access Points (APs) to collect and process vehicle information at a simulated four-way intersection with three lanes per direction. The Priority-based Dynamic algorithm evaluates the traffic demand at each approach at the beginning of each signal cycle, determines the service priority of the approaches, and assigns a green duration based on the number of waiting vehicles, subject to a minimum of 6 seconds and a maximum of 24 seconds. The proposed algorithm was compared with a conventional Fixed-Time algorithm using the same traffic-generation conditions. A total of 275 simulation runs were conducted for each algorithm across four simulation durations. Performance was evaluated using average vehicle waiting time, maximum waiting time, intersection throughput, and fairness.

The results show that the Priority-based Dynamic algorithm consistently outperformed the Fixed-Time algorithm under the simulated conditions. Average vehicle waiting time was reduced by 11.49% to 19.35%, while intersection throughput increased by 12.4% to 16.0% across the tested simulation durations. Maximum waiting time was also reduced, with the

largest observed reduction reaching 18.5%, from 831 seconds under Fixed-Time control to 677 seconds under Priority-based control. The results indicate that adapting signal operation to current traffic demand can reduce vehicle waiting time and improve the number of vehicles processed through the intersection compared with fixed signal timing.

Although the study is limited to a single simulated intersection and controlled traffic-generation conditions, the findings demonstrate the potential of a lightweight VANET-assisted approach for adaptive traffic signal control. The proposed system provides a foundation for further investigation using more complex traffic scenarios, real-world traffic data, multiple interconnected intersections, and more advanced communication and prediction techniques.

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Our thanks extend to the Department of Computer Science and Engineering (CSE) for providing the academic environment, facilities, and guidance necessary to carry out this research. We are particularly grateful to the Chairman Prof. Dr. Utpal Kanti Das sir and Coordinators Shahinur Alam sir and Md Rashedul Islam Rana sir of the CSE Department for their valuable assistance and advice.

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Chapter 1. Introduction

Vehicular Ad Hoc Networks (VANETs) are an important component of Intelligent Transportation Systems (ITS), enabling communication between vehicles (V2V) and between vehicles and roadside infrastructure (V2I). Vehicles operate as mobile nodes within a dynamic wireless network, allowing information such as vehicle position, speed, and traffic conditions to be exchanged for transportation-related applications [1]. VANETs can provide real-time traffic information that can be used for adaptive traffic signal control and intersection management [2], [3]. As cars operate as mobile nodes within a moving wireless network, traffic condition information, car status, and security-related information may be exchanged to allow efficient real-time traffic control and improve road traffic security. VANETs offer higher real-time traffic control performance than static signal systems. They can lessen traffic and delays within densely populated cities. Urban traffic congestion is a significant transportation problem that can increase travel time, fuel consumption, and environmental impacts. Inefficient traffic signal control can further contribute to congestion and delays at signalized intersections [5], [6]. Traditional pre-timed signal-control systems use predetermined timing plans and therefore have limited ability to respond to rapidly changing traffic demand. Adaptive traffic signal control has consequently been investigated as a means of adjusting signal timing according to observed traffic conditions [4], [5], [7]. So they result in inefficient vehicle flow, longer waiting times at intersections, and unnecessary delays.

1.1 Components and Communication in Vehicular Networks

A typical VANET architecture consists of On-Board Units (OBUs installed in vehicles) and Roadside Units (RSUs deployed as fixed infrastructure). These components support vehicle-to-vehicle and vehicle-to-infrastructure communication for transportation applications [1], [8]. In the proposed system, additional Access Points (APs) are introduced as intermediate sensing nodes to collect vehicle information along approaches to the intersection. Together, these technologies support intelligent traffic management by enabling immediate data exchange, vehicle coordination, and adaptive control processes within Intelligent Transportation Systems (ITS).

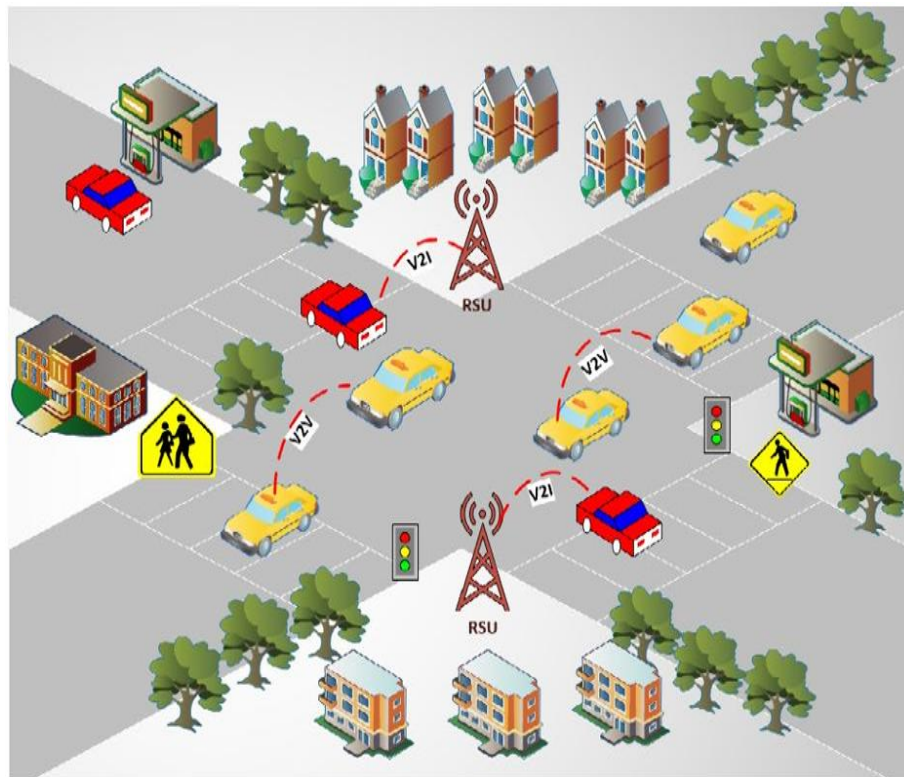


Figure 1.1 Vehicular Network Components and Communications.

Figure 1.1 depicts a multi-tiered vehicular network design and communications. It highlights interaction between vehicles and the base station using V2V and V2I methods,

enhancing smart traffic through efficient interaction between fixed and mobile users. This integrated structure forms the foundation for intelligent traffic management by enabling continuous data exchange between mobile and stationary nodes across the network.

1.1.1 On-Board Unit (OBU):

An On-Board Unit (OBU) is a communication device installed in a vehicle to support wireless communication with other vehicles and roadside infrastructure. Depending on the implementation, vehicle position, speed, and other sensor information can be made available to vehicular-network applications [1], [8].

1.1.2 Road Side Unit (RSU):

Road Side Units (RSUs) serve as fixed infrastructure points that connect mobile nodes within a Vehicular Network. RSUs are tactically deployed at traffic signals, intersections, or roadways. Roadside Units (RSUs) are fixed infrastructure nodes deployed along roads or at locations such as intersections. They provide communication between vehicles and roadside infrastructure and can forward or process traffic-related information for transportation applications [1], [8]. Also, cars will be equipped with enhanced network capability for traffic flows beyond intersections by linking with traffic infrastructures beyond these interfaces (on cloud servers and centralized data centers).

1.1.3 Access Points (APs):

In this study, Aps are introduced as intermediate sensing nodes along each lane leading to an intersection. These units collect real-time vehicle data - such as entry counts, speeds, and directions through short-range communication with passing OBUs (V2I).

Each AP transmits the gathered information to the central RSU (I2I), enabling accurate and timely input for the traffic control algorithm. This distributed sensing approach ensures

scalability and low-latency data flow, making it suitable for complex multi-lane or multi-intersection networks. Access Points can be intelligently used to cover more area and also reduce cost of deploying an RSU.

1.1.4 Communication Modes:

Vehicle communication networks use three primary communication modes:

Vehicle-to-Vehicle (V2V): Allows direct data exchange between vehicles.

Vehicle-to-Infrastructure (V2I): Which allows vehicles to communicate with roadside equipment for monitoring and management.

Infrastructure to Infrastructure (I2I): This facilitates coordination between different infrastructure elements.

There is also hybrid communication, which integrates V2V and V2I, strengthens overall network dependability and adaptability, supporting a wide range of traffic cases and network designs.

1.2 Applications

Over time, vehicle communication networks have expanded from their initial focus on road safety to include a range of applications aimed at making driving more convenient and traffic more efficient [1]. According to experts, Contemporary Intelligent Transportation Systems are widely considered essential because of their ability to integrate communication technology and immediate data processing.

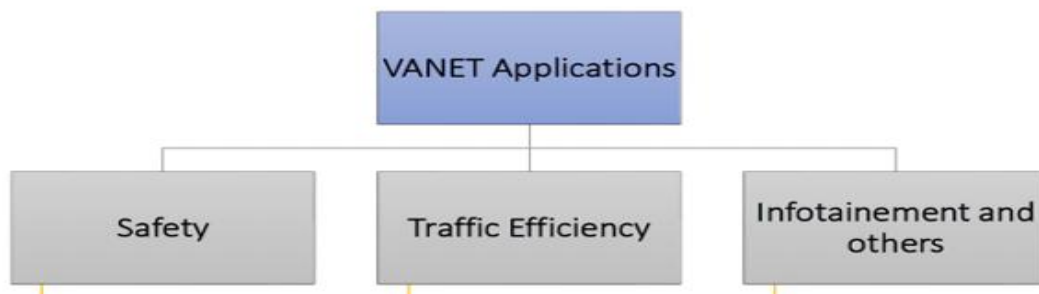


Figure 1.2 Primary Applications of Vehicular Networks.

As illustrated in Figure 1.2, we define 3 application groups that make road transport safer and more reliable, quick and responsive, smart and effective. The applications focused on in the current work are directly concerned with vehicular flow efficiency, as they serve a vital role in the live communication of information for responsive signal control and elimination of traffic congestion.

1.2.1 Safety Applications:

Vehicle Network, the primary purpose of which is focused on safety, especially in reducing or avoiding traffic crashes using risk detection and driver alerting. Enhanced decision-making processes along with timely responses can be supported if vehicles and Roadside Devices can interact and communicate with each other in real-time. The primary applications that will provide this type of benefit will include pre-crash warnings where vehicles behind the car will be alerted when a crash is expected or imminently occurring, alerting vehicle to conditions on the road including holes in the pavement and slippery or icy roads, warning of emergency vehicles indicating that traffic signals for fire trucks or ambulances should be helped to be unblocked or even automatically updated, and collision avoidance in which vehicle data with respect to safety and potential crashes is shared. Such safety precautions

can then enable state-of-the-art applications similar to those that fit the adaptive logic used by the project, including emergency vehicle signal control.

1.2.2 Traffic Efficiency Applications:

VANET-based traffic-efficiency applications include adaptive route guidance, signal timing assistance, traffic-state estimation, and adaptive traffic signal control [1]–[5]. To enhance intersection performance, relieve congestion, and increase overall traffic flow, applications for traffic throughput are invited. Using vehicle data from OBU, AP, and RSU, these apps deliver intelligent decisions in road and signal control. Some important applications include: Adaptive route guidance that warns cars of congested routes in real-time optimization; Green light optimal speed advisory (GLOSA) that notifies cars of impending signal timings to minimize stopping; and dynamic traffic signal control where RSU modifies the duration of green lights in real-time based on the number of vehicles crossing. Likewise, intersection coordination promotes coordinated signal control, while congestion detection allows authorities to manage bottlenecks effectively. These developments show how traffic management can be improved. Further, it can make urban transport eco-friendly and efficient through a VANET-based solution..

1.2.3 Infotainment and Other Services:

Beyond safety and efficiency, Vehicular Networks also enhance driving experience through connectivity and information services. Examples include:

Internet Access and Cloud Connectivity: RSUs can provide internet access to vehicles. So travelers can use internet for free even when on road.

Location Based Services and Advertisements: Businesses can broadcast their offers to the travelling customers through nearby RSUs. So travelers will be motivated to visit the outlets on their way.

Weather and Road Condition Updates: Vehicles can access live data on weather, air quality, or road surface status from Roadside Infrastructure as those will be connected to many cloud servers.

The infotainment applications are not central to this thesis. Nonetheless they represent a growing scope of VANET technology in building connected, intelligent and user centric transportation ecosystems.

1.3 Challenges

Effective management of urban traffic can be achieved using VANETs. However, practical problems primarily associated with RSU and vehicle data for responsive control are hindering real-time performance and large-scale implementation.

1.3.1 Real-Time Data Collection and Processing:

VANET-based systems rely on timely and accurate vehicle data. However, in dense traffic, large volumes of data from multiple vehicles may cause delays or loss during transmission. Real-time processing at RSUs requires high computational resources and efficient prioritization mechanisms. Any delay in data collection can lead to inaccurate traffic signal decisions.

1.3.2 Limited RSU Coverage and High Deployment Cost:

RSUs are critical for vehicle–infrastructure communication, but their high installation and maintenance costs restrict widespread deployment, especially in developing regions. Limited

RSU coverage results in incomplete data collection, reducing the reliability of adaptive control algorithms.

1.3.3 Communication Reliability and Network Congestion:

In busy intersections, simultaneous broadcasting from many vehicles can cause channel congestion and message collisions. Maintaining reliable communication under such conditions is essential for accurate vehicle count estimation and timely signal switching.

1.3.4 Scalability and System Adaptability:

The efficiency of a dynamic control system depends on its capacity to adapt to changing junctions, vehicle densities, and environments. Thus, scaling from one junction to a city network requires coordination and common algorithms. Through VANET-based traffic signal control, road efficiency and well-being can be enhanced. However, high RSU costs, data quality issues, inconsistent data processing, and processing holdups are barriers to adoption.

To summarize the section, we can say VANET based traffic signal control holds immense promise for improving road efficiency and safety, no doubt about it. But there are still several of real world constraints, that hinder its widespread adoption. Limitations such as high RSU deployment costs, inconsistent data communication, and real-time processing delays must be addressed. These are first and foremost necessity to ensure reliable performance. The present research focuses on tackling some of these challenges by developing a system that adapts signal timings based on live vehicle data. We aim to improve the traffic flow efficiency and reduce overall vehicle waiting time through this research.

1.4 Motivation



Figure 1.3 Traffic Jam in Bangladesh.

Figure 1.3 shows a heavy traffic jam in Dhaka City. Traffic jams are a serious issue in the world right now. People lose a lot of time in traffic jams. It increases the fuel consumption. Taking both into consideration a lot of resources and money are wasted for traffic jams. Even in Bangladesh traffic jams are a major issue, costing us a lot of time on the road. People have to spend 2-3 hours extra in traffic jams on the way to their respective destination. Sometimes in major traffic jams this number can become 5/6 hours in a day. A lot of productive work could have been done in this time. One of the main reasons for traffic jams is inefficient traffic signal systems at the intersections. Having an efficient and dynamic traffic signal control system can reduce waiting time and overall traffic jams in major city areas.

The Intelligent Transportation System (ITS) was envisioned to improve road safety and efficiency. Vehicular Ad Hoc Networking (VANET) brings in new dimension to ITS, since

data can be disseminated and shared among cars using On Board Units (OBU) that are installed in vehicles and, or infrastructure services like Road Side Units (RSU) for analytics that can facilitate the different aspects of ITS. By utilizing VANET we can improve the traffic condition and avoid traffic jams. There are many approaches in VANET for tackling traffic jams. One of these approaches is an intelligent traffic signal control system. Having an intelligent traffic signal control system utilizing VANET can reduce the loss of resources in traffic signals and traffic jams. Traditional traffic control systems do not consider real time traffic conditions. A more dynamic need based traffic signal control could improve the overall travel experience of the road users.

1.5 Research Problem

Although many signal control system technologies exist, there are still important management problems at city crossroads. Traditional traffic signal systems often operate on fixed cycles, switching red to yellow to green lights in a predetermined order[5],[7]. It doesn't care about the real-time traffic load from each direction. This leads to inefficiencies such as increased waiting times, fuel consumption, and congestion.

Some modern approaches incorporate machine learning to adapt signal timings. While machine learning is being used for responsive control in intelligent systems, their massive compute and memory requirements limit their deployment due to resource restrictions. Thus, it's important to realize a lightweight traffic signal system that actively determines decisions using real-time traffic information, minimizing vehicle waiting times. In other words, adjust the proportion of green light based on the count of actual traffic vehicles to lessen traffic congestion and achieve an economical control system for the traffic lights.

1.6 Objectives

The primary objectives of this research are:

- I. To design a dynamic traffic signal control strategy using Vehicular Networks.
- II. To utilize RSU and AP to detect vehicles equipped with OBU.
- III. To count vehicles approaching intersection using APs and RSU.
- IV. To utilize the RSU to detect vehicles that cross the intersection.
- V. To determine signal sequence using vehicle load detected from each direction.
- VI. To calculate green signal time of current direction using vehicle count.
- VII. To minimize unnecessary wait time and improve traffic flow efficiency using real-time vehicle data.

1.7 Research Questions

This research aims to answer the following key questions:

1. How can real-time traffic data at crossings be gathered using vehicular networks?
2. To what extent can RSUs and OBUs accurately predict how many cars are approaching or waiting at an intersection?
3. How can the density of vehicles coming from different directions be used to dynamically manage a traffic signal?
4. Which algorithm or method of making decisions is best for choosing the direction of the green signal in real time?

5. In terms of lowering traffic congestion and vehicle wait times, how is the suggested VANET-based dynamic signal control compared against conventional fixed-time or pre-timed traffic signal systems?
6. When VANET data is used for intersection management, what are the communication and time constraints that are faced

1.8 Outline of the paper

The rest of the paper is organized as follows. Section 2 provides a review of related works on traffic signal control strategies, emphasizing VANET-based and dynamic control approaches. Section 3 describes the proposed method for dynamic intersection traffic signal control using RSU and OBU communication, along with details of the simulation setup and implementation using Python and Pygame. Section 4 presents the results and discussion, focusing on performance evaluation and comparisons with traditional fixed-time and recent adaptive signal control methods. Finally, Section 5 concludes the paper and highlights potential directions for future research.

Chapter 2. Literature Review

This section reviews previous research efforts in the field of traffic signal control. Over the years, various approaches have been proposed by researchers to address traffic congestion and optimize signal timing at intersections. These methods range from traditional hardware-based solutions using microcontrollers to more advanced techniques involving Vehicular Ad-hoc Networks (VANETs), Artificial Intelligence (AI), and Machine Learning (ML). In the following subsections, we will discuss 3 most relevant papers related to our research.

A VANET based Intelligent Road Traffic Signalling System - Nafi & Khan (2012). [11]

Intelligent Traffic Control System Based on Round Robin Scheduling Algorithm – Lutfun Nipa & Md. Mohibul Islam (2015). [12]

Design and Implementation of a Smart Traffic Signal Control System for Smart City Applications – Wei-Hsun Lee & Chi-Yi Chiu (2020). [13]

2.1 A VANET based Intelligent Road Traffic Signalling System - Nafi & Khan

In this paper the authors proposed an IRTSS system based on the VANET architecture. They claimed their proposed traffic signalling system can significantly improve traffic flow, energy efficiency and safety of road users. Their methods and final outcomes are discussed in the following sections.

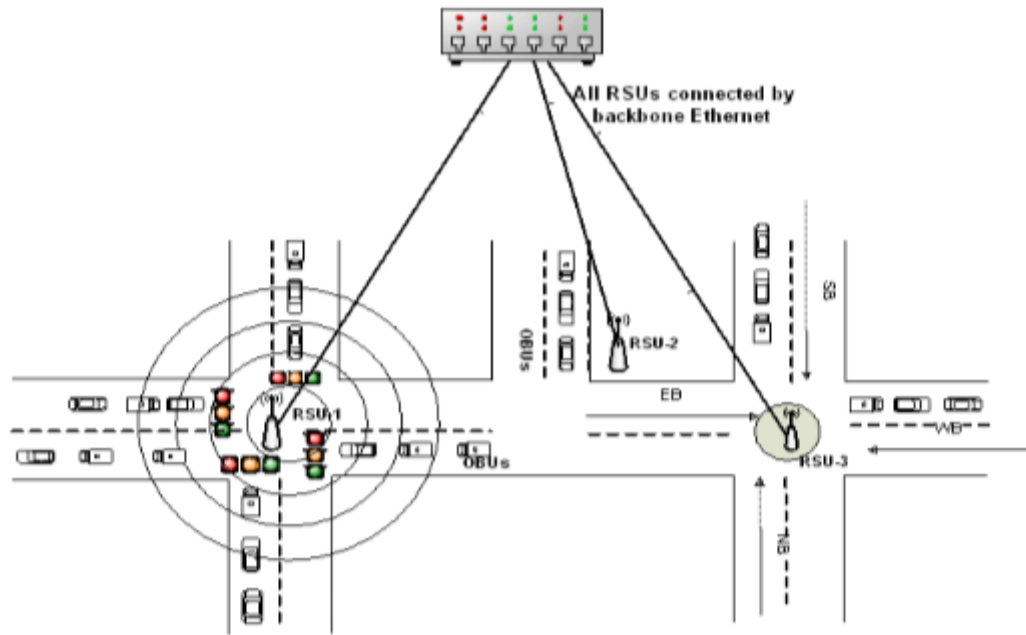


Figure 2.1 System Architecture of Paper-1.

The methodology mainly relies on V2I communication. The On-Board Unit (OBU) of every vehicle periodically sends a unicast message to the nearest RSU. The message contains information about the vehicle such as ID, speed, location (latitude/longitude), and estimated time to reach the intersection. RSUs broadcast signal status updates every second and manage a list of active vehicles updated every 5 seconds.

They have utilized the basic architecture of the V2I communication application of the DSRC consisting of RSUs and OBUs, where RSUs are connected through an ethernet backbone. Each RSU covers 500 meters sized cells that allow all the cars approaching a junction to communicate with the RSU.

The RSU counts vehicles from each direction as East Bound (EB), West Bound (WB), South Bound (SB) and North Bound (NB). Critical lane volumes are computed as Z1 (East-

West) and Z2 (North-South). Green light durations are dynamically calculated using analytical formulas based on vehicle arrival rate, saturation flow rate, startup loss time, safe braking distance. If one direction's traffic exceeds 75% of total volume, a modified green time formula is applied to ensure fairness. OBUs receive signal timing and phase predictions to decide whether to stop or continue driving.

The authors have used a simulation tool called OPNET modeler 17.1 to implement and test their model. They have simulated different traffic densities such as EW: 400 vehicle/hour, NS: 200-400 vehicle/hour. The RSU-OBU communication operated with 6mbps data rate and 1 second broadcast interval.

The authors were able to reduce the average waiting time using their adaptive signaling system compared to a fixed time approach. The fairness was quantitatively measured using Jain's Fairness Index, with values as high as 0.997, which indicates that all directions were treated nearly equal. Their system showed lower standard deviation waiting time across directions contributing to better traffic flow. The maximum waiting time also decreased under the adaptive method compared to fixed timing. Network throughput increased proportionally with traffic volume, while uplink delay remained acceptably low, proving the feasibility of RSU-based data collection even at scale.

	Proposed IRTSS (Avg. waiting time)			Fixed signaling (Avg. waiting time)	
Flow Densities (veh/h)		Max (s)	Standard Deviation	Max (s)	Standard Deviation
<i>EW400-NS400</i>	EB	38	10.30	45	15.17
	WB	42	10.24	45	14.95
	SB	38	10.83	45	14.17
	NB	40	9.65	45	14.59
<i>EW400-NS300</i>	EB	38	9.19	45	15.17
	WB	34	9.12	45	14.95
	SB	37	9.95	45	14.10
	NB	44	10.69	45	15.08
<i>EW400-NS200</i>	EB	36	9.90	45	15.17
	WB	37	9.79	45	14.95
	SB	38	10.53	45	13.70
	NB	34	9.67	45	14.65

Figure 2.2 Result of the system of paper 1.

The system relies on precise analytical models for green/red timing calculation, which may not be scalable or adaptable in all real-world scenarios. In highly imbalanced traffic scenarios, lanes with lower traffic volume may suffer longer delays, despite fairness adjustments. The model assumes full RSU coverage, which may not be affordable or feasible in large or developing cities.

2.2 Intelligent Traffic Control System Based on Round Robin Scheduling Algorithm – Lutfun Nipa & Md. Mohibul Islam

This paper proposed a microcontroller based intelligent traffic control system that applies the Round Robin (RR) scheduling algorithm to regulate signal transitions across a

four way junction. Each road segment is treated as a process (G1 to G4), and the signal controller allows only one gate to pass traffic at a time while others are closed. Signal rotation is managed cyclically, ensuring all directions are served fairly in sequence.

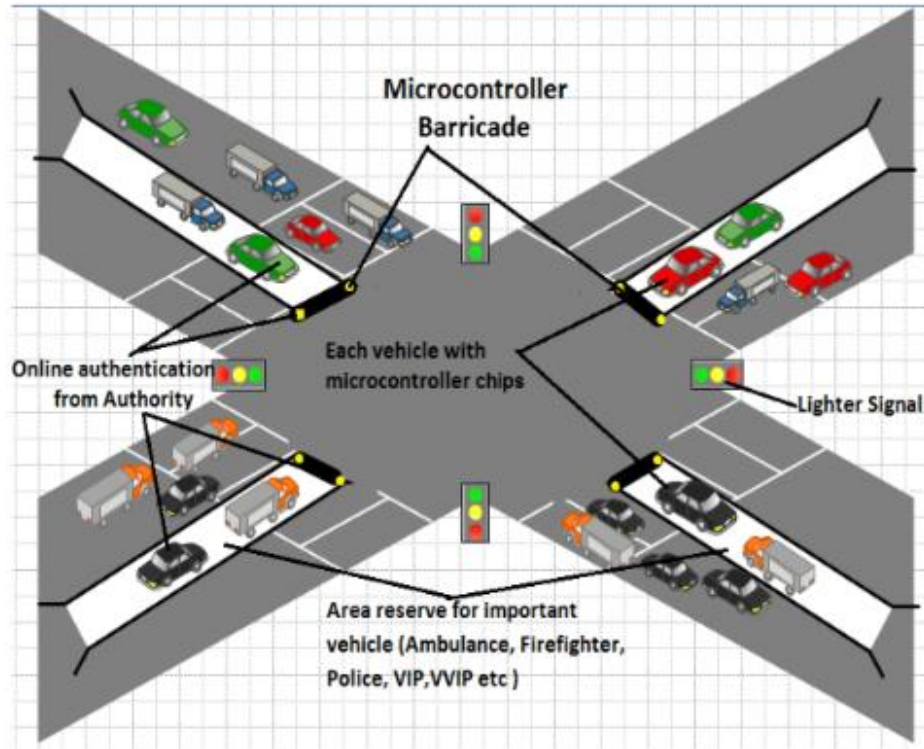


Figure 2.3 Proposed architecture of Paper 2.

The authors modeled the intersection as a 4-gate system, which are G1, G2, G3, and G4. Each of these gates represent a directional lane. A microcontroller opens one gate at a time in a fixed time slice which is 1 minute. Then rotate using a Round Robin like cyclic mechanism. The authors tested 3 different cases depending on arrival time of vehicles. The cases are, different arrival times, all starting at zero and uniform arrival time. Each gate has a service time and burst time. Gantt charts and waiting time calculations are used to analyze performance under each case.

Based on their approach the case where each gate has different arrival time had the lowest average waiting time. The other 2 cases had slightly higher average waits. But overall the Round Robin Based system is shown to be starvation free, treating all gates equally. Simulations indicated the model is practical and potentially applicable to busy junctions like those in Dhaka city.

The system is not adaptive to real-time traffic volume; time slices are fixed regardless of actual vehicle count. No vehicle detection system (like RSUs or sensors) is integrated — the model is purely algorithmic and hardware-based. It assumes all gates have equal importance and fixed quantum time, which may be inefficient in scenarios with highly uneven traffic. It lacks dynamic coordination or learning-based improvements, making it rigid under varying traffic loads.

2.3 Design and Implementation of a Smart Traffic Signal Control System for Smart City Applications – Wei-Hsun Lee & Chi-Yi Chiu

In this paper, the authors developed a smart traffic signal control system (STSC) designed to support various smart city applications. Their system uses roadside units (RSUs) that can communicate with vehicles (via V2X technology) to manage traffic more intelligently. The goal was to improve traffic flow, give priority to emergency and public vehicles, reduce delays, and make the system compatible with existing traffic lights — all while keeping the cost of deployment low.

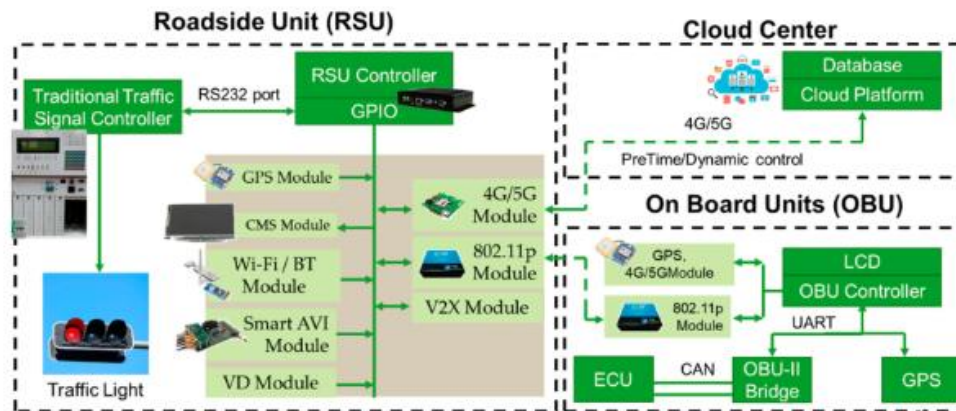


Figure 2.4 System Diagram of Paper 3.

The system is built around RSUs that are installed at intersections and connected to existing signal controllers. These RSUs talk to On-Board Units (OBUs) installed in vehicles using wireless communication. Vehicles broadcast their updates every second. Those broadcasts include their location, speed, direction and type. When an emergency vehicle (like an ambulance) approaches an intersection, the RSU detects it and takes control. After that the RSU either extends the green light. Or if needed cuts the red one to let the vehicle pass quickly and safely. A special blinking light and a display sign also warn nearby drivers and pedestrians that an emergency vehicle is coming. The system was designed in layers with applications, communication modules, and operating systems working together. It can also collect real-time traffic data from detectors or video systems, which can help in adjusting traffic lights adaptively.

The authors built a prototype first. Used Raspberry Pi for the vehicle side (OBU) and an industrial PC for the RSU. And then they tested it both in the lab and in real-world conditions in Tainan, Taiwan. Their tests showed that the system works well especially the emergency vehicle preemption feature. The system responded in real-time and switched traffic signals to

allow emergency vehicles to pass without stopping, improving both traffic flow and safety. The use of blinking lights and message signs made it clear to other road users where the emergency vehicle was coming from. Communication between vehicles and RSUs worked reliably up to a range of 400 meters. The system also handled multiple priority requests smoothly for example, if an ambulance and police car approached at the same time, it would give higher priority to the more urgent one or use a first-come-first-served system if they were equal. The best part is that this system doesn't require cities to rip out and replace their old traffic controllers. This can be installed alongside existing system. So it is easy to adopt.

Despite its strengths, the system still has a few drawbacks. It depends on complete RSU coverage, which might be expensive or impractical for some cities. Also, if communication is interrupted (due to signal loss or hardware issues), the emergency vehicle feature may not work properly.

2.4 Comparison and Discussion of Existing Approaches

The studies reviewed in the previous sections show that traffic signal control can be improved by using information about vehicles approaching an intersection. Traditional scheduling methods such as Round Robin provide a simple way of serving different approaches in a fixed order, while VANET-based systems can use information such as vehicle location, speed, direction, and traffic volume to make more adaptive decisions [11]–[13]. The main difference between these approaches is how much the signal controller responds to the current traffic condition.

Pandit et al. proposed a VANET-based traffic signal control method in which real-time vehicle speed and position information are collected and used to organize vehicles into groups or platoons. The signal control problem was treated as a scheduling problem, and the

oldest-arrival-first approach was used to reduce delay at the intersection [2]. Their results showed that the approach could reduce vehicle delay under light and medium traffic conditions compared with vehicle-actuated, Webster, and pre-timed signal control methods. This demonstrates that vehicle information can be used not only for traffic detection but also for deciding which traffic movement should be served first.

A different approach was presented by Maslekar et al. through the CATS system, which uses car-to-car communication to estimate the density of vehicles approaching an intersection [4]. The estimated density is then used to determine the signal cycle time. Their results showed reductions in waiting time and queue length compared with a traditional pre-timed system and an adaptive fuzzy-logic system. This approach is relevant to the present study because it shows that traffic density obtained through vehicle communication can be directly connected to signal timing decisions.

Connected-vehicle approaches have also been investigated using infrastructure-based communication. Xu proposed an adaptive signal control method that obtains traffic-state information from connected vehicles through roadside units. The method considers information such as elapsed time, vehicle stops, and queue length before adjusting phase sequence and green duration. The study reported lower average delay than an optimized fixed-time control method, including scenarios with relatively low connected-vehicle penetration [7]. Similarly, Nguyen Van Phu et al. developed a V2I-based urban traffic control algorithm in which an RSU builds a map of connected vehicles approaching and leaving an intersection and uses this information to select the traffic phase [15].

These studies indicate that connected-vehicle and VANET technologies provide a more detailed source of traffic information than approaches that depend only on fixed signal

schedules. A general review by Jing et al. found that connected-vehicle information can be used for different adaptive signal-control strategies, while Guo et al. classified a range of connected and automated vehicle approaches according to how traffic states are estimated and how signal timing is controlled [5], [6]. More recent reviews also identify rule-based, mathematical, and artificial-intelligence-based approaches as different directions in connected-vehicle traffic signal optimization [14].

However, greater adaptability also introduces additional requirements. VANETs have highly dynamic network conditions, and changes in topology and connectivity can affect communication between vehicles and infrastructure [8]. Research on VANET communication has identified rapid topology changes and intermittent connectivity as important challenges for reliable data exchange [8]. Therefore, a traffic controller based on vehicle communication should use information that can be obtained reliably and should not depend on unnecessarily complicated communication or prediction mechanisms.

Overall, the reviewed studies demonstrate three important points. First, real-time vehicle information can be used to improve traffic signal decisions [2], [4], [7]. Second, different control strategies can be used to determine which traffic movement receives service, including scheduling, density-based control, and demand-based phase selection [2], [4], [15]. Finally, fairness and reliable service to different traffic movements remain important considerations when one approach has significantly greater traffic demand than another [11], [12], [7]. These observations provide the basis for the approach used in this study.

2.5 Research Gap

Previous research has established that VANET and connected-vehicle technologies can provide useful real-time information for traffic signal control. Existing systems have used

vehicle position and speed [2], traffic density [4], queue-related information [7], and vehicle maps generated through V2I communication [15]. At the same time, scheduling-based methods such as Round Robin provide a simple and predictable mechanism for serving different traffic approaches, but their fixed service pattern does not directly respond to differences in traffic demand [12].

The reviewed studies also differ in their level of complexity. Some approaches use mathematical traffic models or scheduling formulations, while others use vehicle-density estimation, adaptive rules, or more advanced optimization methods [2], [4]–[7]. These approaches demonstrate strong potential, but a practical traffic signal controller does not necessarily require all of these mechanisms. For a smaller-scale implementation, there is value in investigating whether real-time vehicle information can be used with a simpler priority-based decision process.

Another consideration is fairness between traffic approaches. A controller that always selects the approach with the highest number of vehicles may provide good service to heavily congested roads but can repeatedly delay approaches with lower traffic volumes. On the other hand, a completely fixed scheduling method treats every approach similarly even when their traffic demands are very different. Therefore, there is a need for a balance between demand-based priority and fair service to the different approaches.

Based on the reviewed literature, the research gap addressed in this study is the investigation of a relatively simple VANET-assisted traffic signal control mechanism that uses real-time vehicle information to determine traffic priority and adjust green time according to the current demand, while still providing service to competing traffic approaches. The objective is not to replace more complex optimization or learning-based

methods, but to examine whether a lightweight priority-based approach can provide measurable improvements in traffic waiting time and overall intersection performance.

2.6 Relevance to the Proposed System

The findings from the reviewed literature provide the foundation for the design of the proposed traffic signal control system. VANET research shows that vehicles can act as sources of real-time traffic information, while V2I systems demonstrate that roadside infrastructure can collect this information and use it for traffic management [8], [2], [7], [15]. This supports the use of On-Board Units (OBUs), Access Points (APs), and a Roadside Unit (RSU) in the proposed architecture.

Unlike approaches that require detailed traffic prediction or complex optimization, the proposed system focuses on information that can be directly obtained from the vehicles approaching the intersection. The collected vehicle information is used to determine the number of vehicles waiting or approaching from each direction. Based on this information, the controller determines the priority of the traffic approaches and calculates an appropriate green duration.

The proposed approach is also related to the scheduling concept discussed by Nipa and Islam [12] and Pandit et al. [2]. However, instead of following a completely fixed Round Robin sequence or applying a more complex job-scheduling formulation, the proposed controller uses the current traffic demand to influence the selection of the next approach. This allows the signal controller to respond to changes in traffic conditions while retaining a relatively simple control mechanism.

Fairness is also considered in the design. The purpose of using traffic priority is not to allow one direction to occupy the intersection indefinitely. Rather, the priority mechanism is

intended to give additional service to approaches with greater demand while ensuring that other approaches are also served. This consideration is particularly important in an intersection where traffic volumes can differ between directions.

The proposed system therefore combines the main useful ideas identified from the reviewed literature: real-time vehicle information from VANET communication [8], adaptive traffic information processing [2], demand-based traffic estimation [4], and infrastructure-assisted signal control [7], [15]. The resulting system is designed as a comparatively simple priority-based controller that can be implemented and evaluated through simulation.

2.7 Summary

This chapter reviewed existing approaches to traffic signal control, beginning with VANET-based signalling, Round Robin scheduling, and smart traffic signal systems. Additional studies were then examined to provide a broader view of VANET and connected-vehicle-based traffic control. The literature shows that real-time information from connected vehicles can be used to estimate traffic conditions and improve signal decisions. Different researchers have approached the problem using vehicle scheduling, traffic-density estimation, queue information, and V2I-based traffic maps [2], [4], [7], [15]. Review studies also show that connected-vehicle traffic control is a broad research area with several possible control and optimization strategies [5], [6], [14].

At the same time, communication reliability, system complexity, and fairness between traffic movements remain important considerations [8], [12]. Based on these observations, this study focuses on a lightweight priority-based traffic signal controller that uses real-time vehicle information to determine traffic demand and dynamically allocate signal service. The following chapter describes the methodology and implementation of the proposed system.

Chapter 3. Research Methodology

The chapter presents the design, implementation, and experimental evaluation of the proposed VANET-based dynamic traffic signal control system. The methodology was developed to investigate whether real-time vehicle information can be used to allocate traffic signal priority and green-light duration according to the traffic demand at a four-way intersection. The proposed system is evaluated against a conventional fixed-time traffic signal control method under the same simulation conditions.

The overall methodology consists of four main stages: (1) designing the VANET-based system architecture, (2) developing a simulation model of the intersection and vehicle movement, (3) implementing and executing the fixed-time and priority-based dynamic traffic signal control algorithms, and (4) collecting and analyzing simulation data using predefined performance metrics. Python and Pygame were used to implement and visualize the simulation, while vehicle and signal events were logged for subsequent quantitative analysis.

3.1 System Architecture and Requirements

The proposed system is designed around Vehicle-to-Infrastructure (V2I) communication, in which vehicles equipped with On-Board Units (OBUs) provide traffic information to roadside infrastructure [17]. The architecture consists of four principal components: On-Board Units (OBUs), Access Points (APs), a Road Side Unit (RSU), and a Traffic Signal Controller (TSC). Together, these components form the communication and control framework used to obtain traffic information and make signal-control decisions.

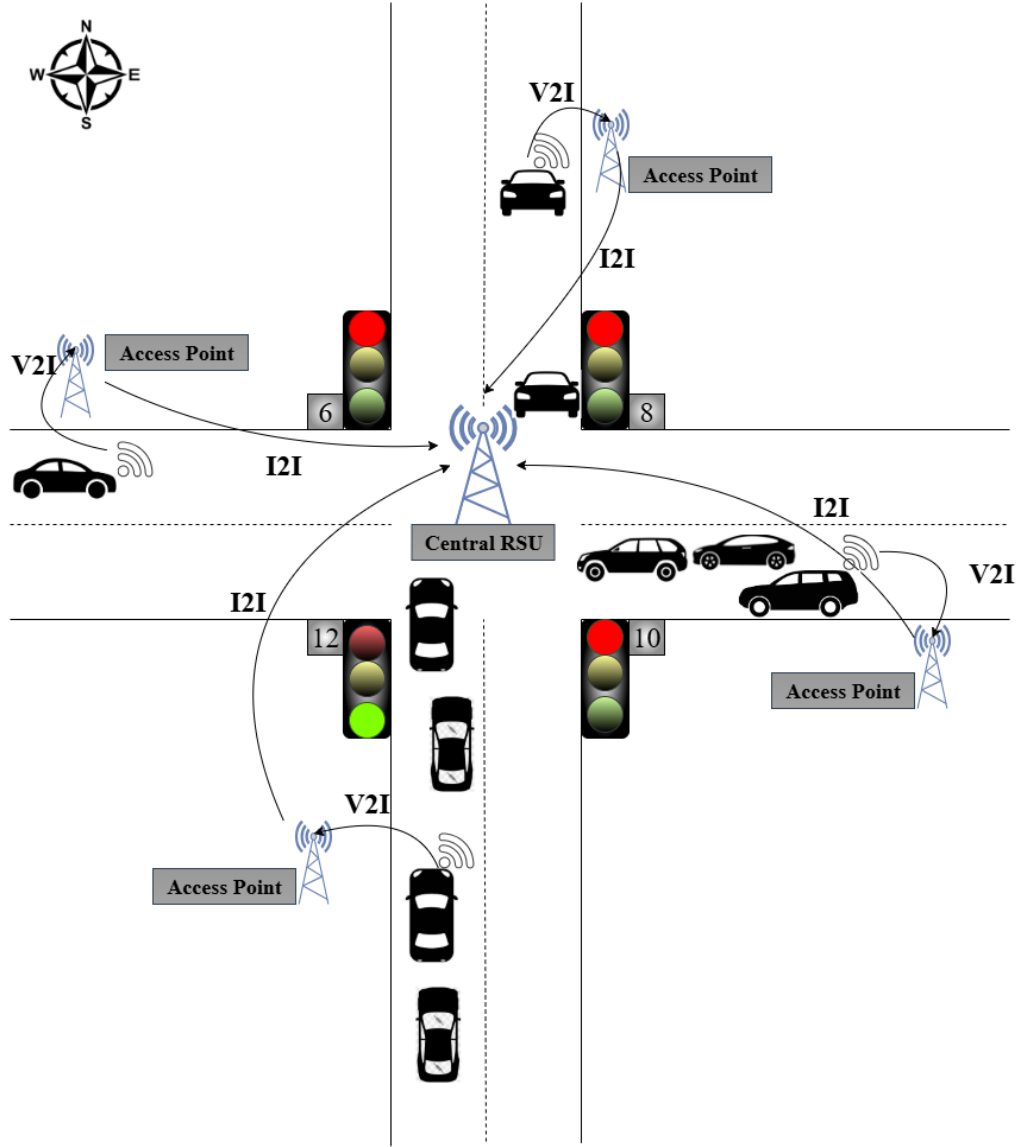


Figure 3.1 Proposed architecture.

Figure 3.1 illustrates the communication and decision-making process of the proposed system. The process begins with vehicles approaching a signalized intersection. Each vehicle is equipped with an OBU that represents the vehicle's communication capability and provides information such as vehicle ID, vehicle type, speed, direction, and position. This information represents the traffic state that can be observed by the roadside infrastructure.

As vehicles approach the intersection, their information is received by Access Points (APs) positioned along the approaches. The APs act as distributed data-collection points and forward the collected vehicle information to the central Road Side Unit (RSU). This arrangement allows traffic information from multiple approaches to be gathered and processed at a central location rather than relying on a single sensing point.

The RSU functions as the primary decision-making component of the system. It aggregates the information received from the APs and determines the number of vehicles waiting on each approach. These vehicle counts are then used by the priority-based dynamic traffic signal control algorithm to determine which direction should be served next. The direction with the higher current traffic demand receives higher priority.

After selecting the direction to be served, the RSU determines the green-light duration according to the number of vehicles on that approach. The calculated green duration is sent to the Traffic Signal Controller (TSC), which applies the corresponding signal state at the intersection. Importantly, the green duration is calculated immediately before a direction is served and remains fixed for the entire green phase. New vehicles may arrive while the phase is active, but they do not change the duration of the ongoing green phase. These vehicles are considered during the next signal decision.

The complete control process therefore follows a feedback-based sequence. Vehicle information is generated by OBUs, collected by APs, forwarded to the RSU, processed to determine traffic demand, and used to select the next signal phase. The TSC then executes the selected signal phase. Once the phase is completed, the traffic condition is evaluated again and a new signal decision is made. This process allows the proposed controller to

respond to changes in traffic demand while maintaining a predictable signal duration during an active green phase.

The architecture was implemented as a simulation model rather than a physical VANET deployment. The simulation reproduces the interaction between vehicles and roadside infrastructure while allowing the proposed signal-control strategy to be compared with a conventional fixed-time strategy under controlled conditions. The same intersection layout, vehicle-generation mechanism, vehicle characteristics, and environmental parameters are used for both algorithms to ensure a consistent basis for comparison.

3.2 Proposed System Architecture

The proposed system is designed as a VANET-assisted traffic signal control framework in which vehicle information is collected through roadside communication infrastructure and used to make traffic signal decisions. The architecture consists of four main components: On-Board Units (OBUs), Access Points (APs), a Road Side Unit (RSU), and a Traffic Signal Controller (TSC). Their interaction enables the system to observe traffic conditions at the intersection and dynamically determine which approach should receive the next green phase.

As illustrated in Figure 3.1, each simulated vehicle is equipped with an OBU that represents its communication capability and provides information associated with the vehicle, including its identifier, vehicle type, position, speed, and travel direction [17]. Vehicles are generated continuously at a fixed rate of 5 vehicles per second. When a vehicle is generated, its approach direction is selected randomly, with each of the four approaches having an equal probability of 25%. Consequently, individual simulation runs may contain different traffic distributions among the four approaches even though the expected

probability is equal over a large number of vehicles. Vehicle type is also selected randomly from the four modeled types: car, bus, truck, and bike.

The APs are positioned along the approaches to the intersection and represent the roadside data-collection layer. They receive information from vehicles approaching the intersection and provide the traffic information required by the central controller. The RSU aggregates the information collected from the approaches and maintains the current traffic state of the intersection. In particular, the number of vehicles waiting on each approach is used by the priority-based control algorithm to determine the next direction to be served.

The RSU performs the primary signal-control decision. At each decision point, the current traffic information is evaluated and the approaches are ranked according to their traffic demand. The approach with the highest priority is selected for service. The green-light duration for the selected approach is then calculated according to the implemented dynamic green-time rule:

$$\text{Green Time} = \min(\max(6, \text{Vehicle Count} \times 0.75), 24) \text{ seconds}$$

The coefficient of 0.75 seconds per vehicle was established through repeated simulation trials during the development of the proposed system. The resulting green duration is constrained to a minimum of 6 seconds and a maximum of 24 seconds. These limits prevent the calculated green phase from becoming excessively short or excessively long.

An important characteristic of the proposed control mechanism is that the green duration is determined once immediately before the selected approach enters the green phase. Once the green phase begins, its duration remains fixed until the phase is completed. Vehicles that arrive during an active green phase do not modify the duration of that ongoing phase. Their presence is considered during the subsequent signal-control decision. This design provides a

predictable signal phase while still allowing the controller to respond to changing traffic conditions between phases.

After the selected green phase ends, a fixed 4-second yellow phase is applied before control is transferred to the next approach. The RSU then reassesses the current vehicle information and determines the next priority order. Therefore, the signal sequence is not predetermined under the proposed method; it is determined from the traffic condition observed at each decision point.

The TSC receives the signal-control decision from the RSU and executes the corresponding signal state. This separates the decision-making process from signal actuation: the RSU determines which approach should be served and for how long, while the TSC applies the resulting signal phase to the intersection.

The overall operation can therefore be summarized as a repeated feedback process: vehicles generate traffic information, APs collect the information, the RSU evaluates the current traffic demand and selects the next approach, the green duration is calculated and fixed for that phase, and the TSC executes the signal change. After the phase is completed, the process is repeated using the updated traffic state. This architecture provides the basis for comparing the proposed priority-based dynamic controller with the fixed-time controller under the same simulated traffic environment.

3.3 Simulation Model and Environment

This section describes the design and execution of the simulation model developed to evaluate the effectiveness of a dynamic traffic signal control algorithm. The simulation models vehicle movement, traffic signal operations, and Vehicle-to-Infrastructure (V2I) communication within a VANET-based four-way signalized intersection [18]. The primary

objective is to compare the proposed priority-based dynamic traffic signal control algorithm with a conventional fixed-time signal control algorithm under the same traffic-generation conditions. The simulation records vehicle and signal events to support quantitative performance analysis. The model was developed using Python and the Pygame library, providing both the simulation logic and graphical representation of the traffic environment.

The simulation follows a modular structure in which the traffic environment, vehicle behavior, communication components, and signal-control algorithms operate as separate components. This structure allows the same traffic environment and vehicle-generation mechanism to be used when evaluating different signal-control strategies. Consequently, changes in performance can be attributed primarily to the signal-control algorithm rather than differences in the underlying simulation environment.

3.3.1 Simulation Environment Overview:

The simulation represents a four-way signalized intersection with four incoming approaches and four corresponding outgoing approaches. Each approach is associated with a traffic signal and a vehicle queue. Vehicles are generated continuously during a simulation run at a fixed rate of 5 vehicles per second. The vehicle generation rate remains constant throughout the simulation and is not changed according to traffic conditions or simulation time.

For every generated vehicle, the travel direction is selected randomly from the four available approaches. The direction-selection mechanism provides an equal probability of 25% for each direction. Therefore, the simulation does not intentionally favour any particular approach during vehicle generation. Vehicle type is also selected randomly from the available vehicle categories: car, bike, bus, and truck.

The simulation uses a discrete time-step mechanism to update vehicle states, traffic signals, and queues. Vehicles move through the intersection when permitted by the active signal and stop when required by a red signal or by another stopped vehicle ahead. Waiting time is accumulated while a vehicle remains stopped and is recorded when the vehicle subsequently resumes movement.

An event-logging mechanism records relevant vehicle and signal information during each run. These records are subsequently used for quantitative comparison of the fixed-time and priority-based control algorithms.

Key features of the simulation environment:

- Real-time graphical visualization using Pygame.
- A synchronized time-step mechanism for vehicle and signal updates.
- Constant vehicle generation at 5 vehicles per second.
- Random selection of vehicle direction with 25% probability per direction.
- Random selection of vehicle type.
- Vehicle and signal level event logging.
- FIFO based queue management.
- Modular implementation supporting multiple traffic signal control algorithms.

3.3.2 Intersection and Vehicle Modelling:

A four-way signalized intersection is modeled with four incoming approaches: Right, Down, Left, and Up. Each approach has its own traffic signal and vehicle queue. Vehicles are modeled as individual entities with attributes including a unique vehicle ID, vehicle type, travel direction, position, entry time, and waiting time.

The available vehicle types are car, bike, bus, and truck. The vehicle type of each newly generated vehicle is selected randomly. Similarly, the travel direction is randomly selected from the four approaches, with each direction having a probability of 25%. This randomization is applied independently to generated vehicles and is kept consistent across the compared algorithms.

Vehicles move through their assigned approach toward the intersection. When the corresponding signal is red, vehicles stop at the appropriate stopping position and remain in the queue until movement is permitted. Vehicles proceed through the intersection when the signal becomes green. Queue processing follows a First-In-First-Out (FIFO) principle, ensuring that vehicles within the same approach are served according to their arrival order.

Vehicle movement speed is kept consistent in the experimental environment rather than being assigned different movement speeds according to vehicle type. The traffic-generation rate is also fixed at 5 vehicles per second throughout every simulation run. Therefore, vehicle type does not introduce an additional speed-based experimental variable.

The simulation maintains the state of each vehicle throughout its traversal of the intersection. This makes it possible to measure individual waiting times and determine when vehicles successfully complete their traversal. These individual records form the basis for calculating the aggregate performance metrics used in the experimental evaluation.

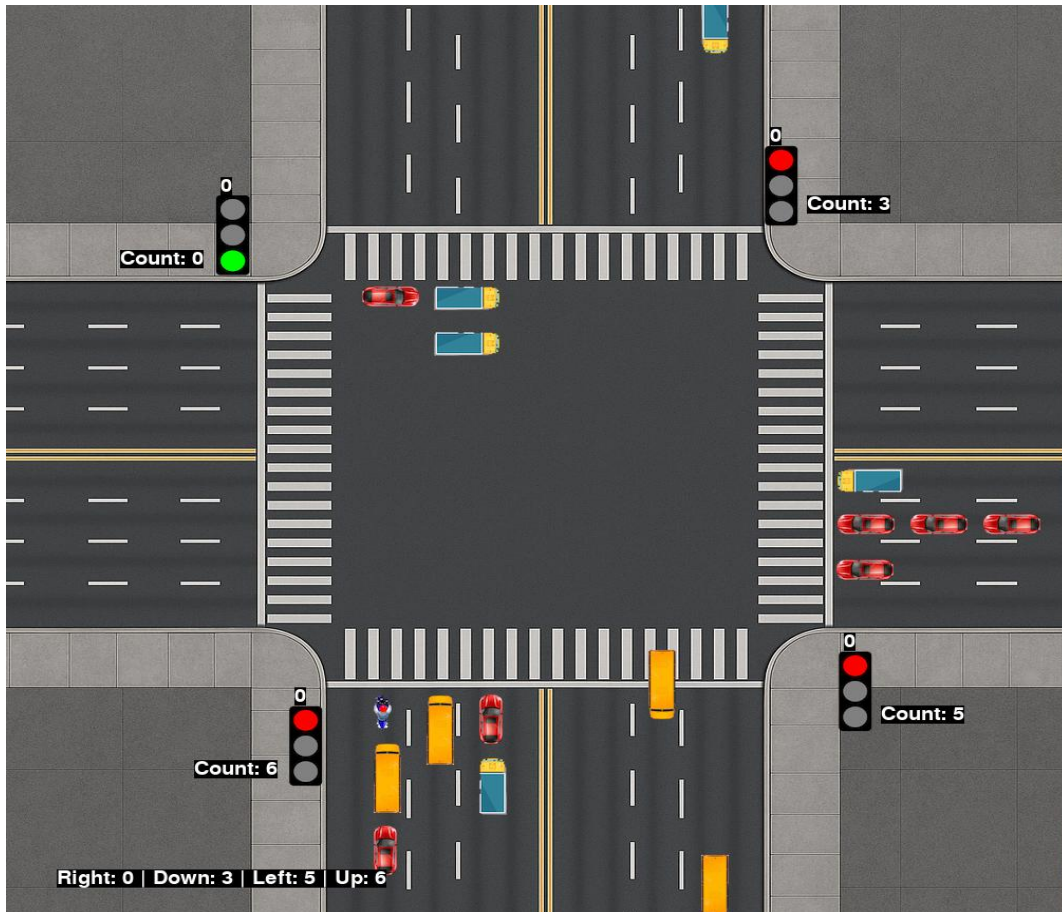


Figure 3.2 Simulation Image.

Figure 3.2 illustrates the implemented simulation environment developed using Python and Pygame. The graphical environment represents the four-way intersection, approaching vehicles, traffic signals, and vehicle movement through the intersection.

3.3.3 Simulation Parameters:

The principal simulation parameters used in the experiments are summarized in Table 3.1. These parameters define the intersection configuration, vehicle-generation mechanism, signal timing, and data recorded during simulation.

Table 3.1: Simulation configuration and parameters.

Parameter	Description	Value/Range
Intersection	Signalized intersection modeled in the simulation	Four-way
Approaches	Incoming traffic directions	Right, Down, Left, Up
Lanes	Lanes available for each direction	3
Vehicle types	Vehicle categories represented	Car, Bike, Bus, Truck
Vehicle type selection	Method used to determine vehicle type	Random
Direction selection	Method used to determine vehicle direction	Random
Direction probability	Probability assigned to each direction	25% per direction
Vehicle generation rate	Number of vehicles generated per second	5 vehicles/second
Vehicle speed	Movement speed used in the simulation	Consistent across vehicles
Fixed-time green duration	Green phase in the baseline algorithm	12 seconds
Dynamic green duration	Green phase range in the proposed algorithm	6–24 seconds
Yellow phase duration	Duration between consecutive green phases	4 seconds

The parameters in Table 3.1 are held constant across the compared algorithms wherever applicable. In particular, the vehicle generation rate, random vehicle-generation mechanism, intersection configuration, vehicle categories, lane configuration, and vehicle movement conditions are not modified when switching between the fixed-time and priority-based algorithms.

Each experimental run uses a predefined simulation duration of 5, 10, 20, or 30 minutes. The simulation terminates when the specified duration is reached, regardless of the number of vehicles generated or processed during the run. The same duration settings and traffic-generation conditions are applied to both algorithms, allowing their performance to be evaluated under equivalent experimental conditions.

The developed simulation therefore provides a controlled and repeatable environment for evaluating adaptive traffic signal control. Its graphical representation enables direct observation of traffic behavior, while the event-logging mechanism provides the data required for quantitative performance analysis.

3.4 Simulation Algorithms and Experimental Setup

This section describes the traffic signal control algorithms implemented in the simulation and the experimental procedure used to compare their performance. Two control strategies are evaluated: a conventional Fixed-Time Algorithm, which serves as the baseline, and the proposed Priority-Based Dynamic Algorithm, which adjusts signal operation according to vehicle information collected from the intersection.

Both algorithms operate within the same simulation environment. The vehicle-generation mechanism, vehicle types, direction-selection mechanism, movement conditions,

intersection configuration, and experimental vehicle count remain consistent so that the comparison focuses on the effect of the signal-control strategy.

3.4.1 Traffic Signal Control Algorithms:

Two traffic signal control algorithms were implemented: the Fixed-Time Algorithm and the Priority-Based Dynamic Algorithm.

Fixed-Time Algorithm

The fixed-time algorithm represents a conventional traffic signal control strategy in which signal timing is predetermined and does not depend on the number of vehicles waiting at the intersection.

The four directions are served sequentially according to the fixed order:

Down → Left → Up → Right

The sequence repeats continuously throughout the simulation. Each direction receives a fixed green phase of 12 seconds, followed by a 4-second yellow phase before the signal changes to the next direction.

Because the green duration and service order are predetermined, the algorithm does not adapt its operation according to the current queue sizes. A direction with relatively few vehicles may therefore receive the same green duration as a direction containing a larger queue.

The fixed-time algorithm serves as the baseline against which the proposed priority-based dynamic algorithm is evaluated.

Priority-Based Dynamic Algorithm

The priority-based dynamic algorithm uses vehicle information collected through the Access Points (APs) and Roadside Unit (RSU) to determine which direction should receive service and how long its green phase should remain active [19].

At the beginning of each cycle, the RSU obtains the current vehicle information from the four approaches. The waiting vehicles are then evaluated according to the traffic information collected for each direction. The directions are ordered according to the calculated priority, allowing the direction with the greater traffic demand to be served earlier.

The green duration is calculated using the following expression:

$$\text{Green Time} = \min(\max(6, \text{vehicle_count} \times 0.75), 24) \text{ seconds}$$

The resulting green duration is constrained between a minimum of 6 seconds and a maximum of 24 seconds. The value of 0.75 seconds represents the estimated time required for one vehicle to pass through the intersection under the simulation conditions. This value was determined through repeated simulation trials and is treated as a fixed model parameter during the experiments.

After a direction has been served, the algorithm proceeds to the next direction according to the priority order. At the beginning of a subsequent cycle, the current vehicle information is evaluated again and a new priority order can be generated. Therefore, the service order is not permanently fixed; it can change between cycles according to the traffic information available at that point in the simulation.

This mechanism allows directions with greater traffic demand to receive longer green phases while maintaining a minimum service duration for other directions.

3.4.2 Simulation Configuration:

The experiments are designed as controlled comparisons between the fixed-time and priority-based algorithms. Both algorithms are executed using the same environmental configuration and vehicle-generation mechanism.

A key requirement of the experimental design is that the traffic demand must remain consistent between the algorithms. Therefore, the vehicle generation rate is maintained at 5 vehicles per second throughout each simulation. Direction and vehicle type are generated randomly using the same generation mechanism, with each direction having a 25% probability of being selected.

The experimental comparison therefore focuses on how effectively each algorithm manages traffic over the same period. For each duration, multiple simulation runs are conducted and the resulting performance measures are recorded. The mean values across the runs are then calculated to provide a representative result for each algorithm at each simulation duration. This allows changes in waiting time, traffic flow, and intersection performance to be compared across different experimental periods.

Table 3.2: Experimental design and simulation protocol.

Parameter	Value / Setting
Algorithms compared	Fixed-Time vs. Priority-Based Dynamic
Vehicle generation rate	5 vehicles/second
Vehicle direction	Random
Direction probability	25% per direction
Vehicle type	Random
Vehicle types	Car, Bike, Bus, Truck
Intersection	Four-way
Lanes per direction	3
Fixed green phase	12 seconds
Dynamic green phase	6–24 seconds
Yellow phase	4 seconds
Vehicle count per experiment	Fixed

Parameter	Value / Setting
Termination condition	When the specified time limit is reached
Randomness	Fresh random generation for vehicle attributes
Primary comparison	Efficiency of processing the same traffic demand

The fixed vehicle-count approach also prevents the comparison from being affected by different numbers of vehicles being completed within an arbitrary time limit. Both algorithms are therefore run multiple times so the mean number of vehicles processed and the workload are similar for both.

3.4.3 Data Logging and Processing:

During every simulation run, detailed event logs are maintained to capture the behavior of vehicles and the operation of the traffic signals. Each vehicle record contains information such as timestamp, vehicle ID, vehicle type, direction, and accumulated waiting time.

Signal events are also recorded, including signal transitions, timestamps, and the direction receiving the green phase. These records make it possible to reconstruct the sequence of signal operations and relate vehicle waiting behavior to signal decisions.

The collected records are exported in CSV format for post-simulation processing. The data from the fixed-time and priority-based algorithms are kept separately so that the results can be aggregated and compared after all experimental runs have been completed.

The logging process is performed during the simulation rather than requiring intermediate recalculation or modification of the experimental parameters. The traffic-generation rate and other predefined environmental parameters remain fixed throughout a run.

3.4.4 Performance Evaluation Metrics:

Several quantitative metrics are used to evaluate the performance of the two traffic signal control algorithms [20]. These metrics measure vehicle delay, congestion, and the ability of each algorithm to process traffic through the intersection.

Average Vehicle Wait Time (AVWT):

Average Vehicle Wait Time represents the mean accumulated waiting time experienced by vehicles during their traversal of the intersection. Waiting time is accumulated whenever a vehicle is stopped because of a red signal or another stopped vehicle.

For a simulation run, the average vehicle waiting time is calculated as:

$$\text{AVWT} = \Sigma \text{ Total Vehicle Waiting Time} / \text{Number of Vehicles}$$

When multiple independent runs are performed for an algorithm, the resulting run-level measurements can subsequently be aggregated to obtain the overall performance of that algorithm.

Maximum Wait Time (MWT):

Maximum Wait Time represents the highest accumulated waiting time experienced by an individual vehicle during a simulation run. This metric is useful for identifying severe delays and potential starvation of vehicles on particular approaches.

A lower maximum waiting time indicates that the algorithm is better able to prevent individual vehicles from experiencing excessively long delays.

Intersection Throughput (IT):

Intersection Throughput represents the number of vehicles that successfully complete their traversal of the intersection during the evaluated simulation. Higher throughput indicates that the signal-control strategy is able to process traffic more efficiently.

Because the experimental design uses a fixed number of vehicles, throughput can also be examined together with the time required to process the complete vehicle set. This provides an additional measure of how efficiently each algorithm handles the same traffic demand.

Simulation Completion Time:

Simulation Completion Time represents the elapsed simulation time required for the specified number of vehicles to be processed and recorded. Since every experimental run contains the same predefined vehicle workload, a lower completion time indicates that the corresponding signal-control strategy processed the traffic demand more efficiently.

Together, AVWT, MWT, IT, and Simulation Completion Time provide complementary measures of algorithm performance. Waiting-time metrics evaluate vehicle-level delay, throughput evaluates the amount of traffic successfully processed, and completion time evaluates the overall efficiency of processing the same predefined traffic demand.

Chapter 4. Results and Discussion

This chapter presents the results obtained from the experiments described in Section 3.3 and compares the performance of the proposed Priority-based Dynamic Traffic Signal Control algorithm with the conventional Fixed-Time control algorithm. To obtain more reliable results, multiple independent simulation runs were conducted for each simulation duration. A total of 150 runs were performed for the 5-minute simulations, 80 runs for the 10-minute simulations, 30 runs for the 20-minute simulations, and 15 runs for the 30-minute simulations. This resulted in 275 simulation runs for each algorithm.

Both algorithms were evaluated under the same traffic-generation conditions. Vehicles were generated at a constant rate of 5 vehicles per second, while the direction of an incoming vehicle was determined through the random generation mechanism described in Section 3.3. Vehicle type was also selected randomly during the simulation. Keeping these conditions identical for both algorithms ensured that the observed differences were primarily associated with the signal-control strategies rather than differences in traffic input.

The performance of the two algorithms was evaluated using four main metrics: Average Vehicle Waiting Time (AVWT), Maximum Waiting Time (MWT), Intersection Throughput (IT), and Fairness. These metrics were selected to evaluate both the overall efficiency of the intersection and the distribution of waiting time among vehicles and approaches. The results for each metric are presented in the following sections, followed by an overall discussion of the observed performance.

4.1 Average Waiting Time

AVWT Average Vehicle Waiting Time (AVWT) was used as one of the primary indicators of traffic efficiency. A lower waiting time indicates that vehicles are able to pass through the intersection with less delay.

In this study, waiting time refers specifically to the period during which a vehicle remains stationary because of the traffic signal or vehicles queued ahead of it. The waiting-time measurement begins when the vehicle stops and ends when it resumes movement and crosses the intersection. Time spent slowing down or moving at a reduced speed because of traffic congestion is not included in the waiting-time measurement. Therefore, the metric represents the time during which vehicles are actually halted rather than their total travel or delay time.

Table 4.1 Average Vehicle Wait Time per run.

Simulation Duration	Fixed-Time Waiting Time (s)	Priority-Based Waiting Time (s)	Wait Reduction
5 Mins	65.9	58.33	11.49%
10 Mins	122.55	102.65	16.24%
20 Mins	233.89	189.8	18.85%
30 Mins	342.43	276.16	19.35%

As shown in Table 4.1, the average waiting time increased for both algorithms as the simulation duration increased. This is expected because vehicles continue to enter the intersection throughout the simulation, resulting in greater traffic accumulation over longer periods. However, the Priority-based algorithm consistently produced lower waiting times than the Fixed-Time algorithm.

For the 5-minute simulations, the average waiting time decreased from 65.90 seconds with the Fixed-Time algorithm to 58.33 seconds with the Priority-based algorithm, representing an improvement of 11.49%. The reduction increased to 16.24% for the 10-minute simulations and 18.85% for the 20-minute simulations. For the 30-minute simulations, the Priority-based algorithm reduced the average waiting time by 19.35%, from 342.43 seconds to 276.16 seconds.

These results indicate that the advantage of the Priority-based algorithm becomes more noticeable as the simulation continues. The Fixed-Time algorithm maintains predetermined signal timings regardless of the traffic conditions at the approaches. In contrast, the Priority-based algorithm uses the observed traffic conditions to determine which approach should receive priority and adjust the signal timing accordingly. This allows available green time to be used more effectively when traffic becomes unevenly distributed among the approaches.

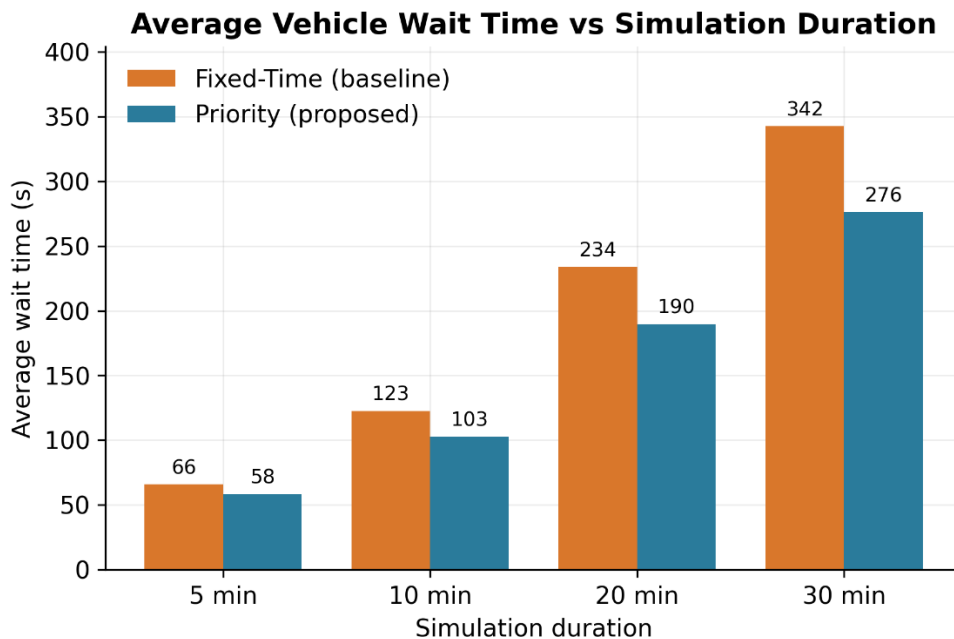


Figure 4.1 Average Vehicle Waiting Time Reduction in comparison to Fixed method.

Figure 4.1 presents the % decrease in waiting time as simulation duration rises. Reductions of 11.5%, 16.2%, 18.9%, and 19.4% were attained by the priority-based controller for 5, 10, 20, and 30 minutes, respectively. Between five and twenty minutes, performance increases dramatically before settling at about 19%. It is suggested that, rather than providing only temporary gains, adaptive signal control consistently outperforms the Fixed-Time approach while retaining efficiency during extended periods of high traffic.

4.2 Maximum Waiting Time

Maximum Waiting Time (MWT) represents the longest waiting time experienced by an individual vehicle during a simulation. Unlike AVWT, which represents the overall average, MWT provides an indication of the worst-case waiting condition experienced by a vehicle.

The maximum waiting time was determined from the recorded vehicle waiting times across the simulation runs for each corresponding simulation duration. This metric is particularly important because an algorithm may achieve a relatively low average waiting time while still allowing some vehicles to remain in the queue for a considerably longer period.

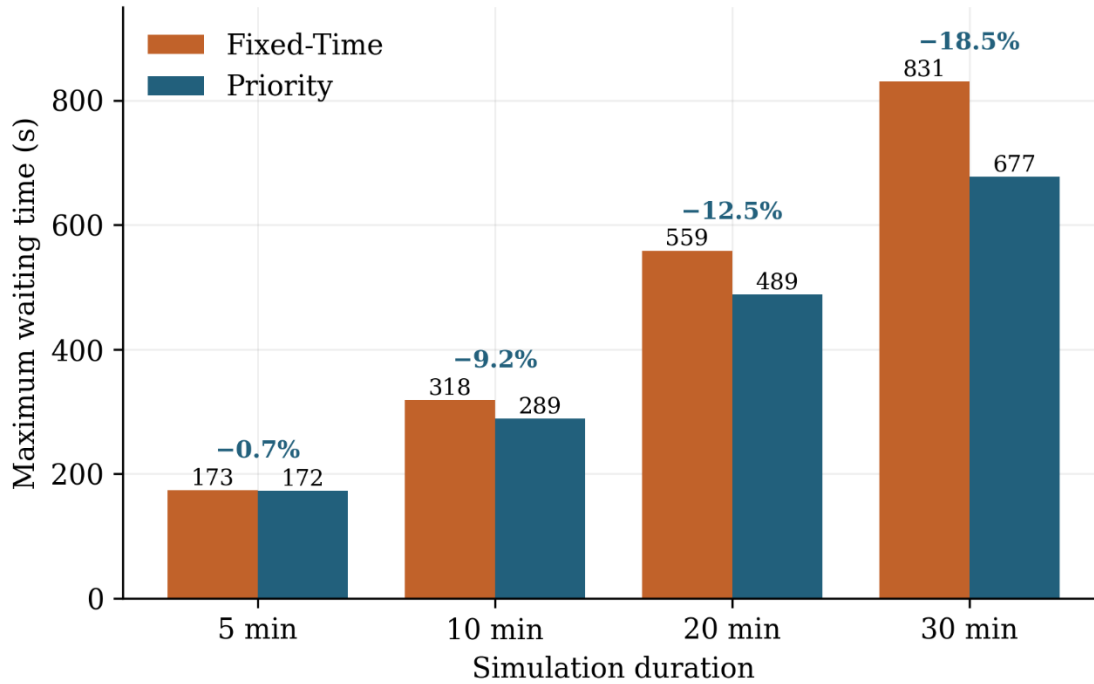


Figure 4.2 Maximum Waiting Time Among all runs.

Figure 4.2 shows the maximum waiting time observed for the Fixed-Time and Priority-based algorithms at different simulation durations. The maximum waiting time increased as the simulation duration became longer for both algorithms. However, the difference between the two approaches became more noticeable at longer simulation durations.

For the 5-minute simulations, the maximum waiting time was approximately the same for both algorithms, resulting in only a 0.7% reduction with the Priority-based approach. The reduction became more noticeable for longer simulations, reaching 9.2% at 10 minutes, 12.5% at 20 minutes, and 18.5% at 30 minutes.

The highest maximum waiting time recorded during the experiments was 831 seconds for the Fixed-Time algorithm. Under the same 30-minute simulation condition, the Priority-based algorithm recorded a maximum waiting time of 677 seconds. This represents a reduction of 154 seconds in the worst-case waiting time.

The results indicate that the Priority-based approach does not only reduce the average waiting time but also reduces the severity of the longest individual delays [15]. This becomes particularly evident as the traffic demand continues over a longer period.

4.3 Intersection Throughput

Intersection Throughput (IT) represents the number of vehicles that successfully cross the intersection within a given period. In this study, throughput is expressed as vehicles served per minute and is used to evaluate the ability of each signal-control strategy to process incoming traffic.

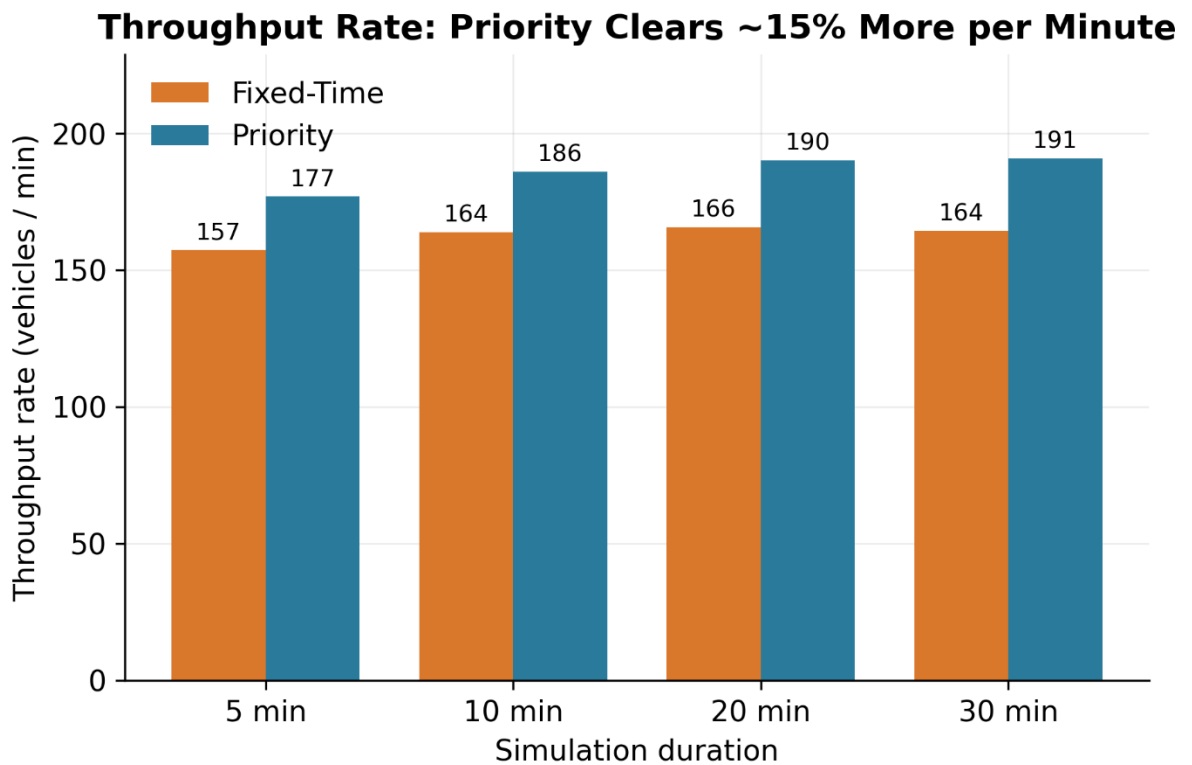


Figure 4.3 Average throughput per run (vehicles served).

Figure 4.3 presents the average throughput obtained by the Fixed-Time and Priority-based algorithms for the four simulation durations. The Priority-based algorithm achieved higher throughput than the Fixed-Time algorithm in all four cases.

For the 5-minute simulations, the Fixed-Time algorithm processed an average of 157.44 vehicles per minute, while the Priority-based algorithm processed 176.96 vehicles per minute. This represents a 12.4% improvement in throughput.

For the 10-minute simulations, the throughput increased to 163.80 vehicles per minute for the Fixed-Time algorithm and 186.07 vehicles per minute for the Priority-based algorithm, resulting in a 13.6% improvement. At 20 minutes, the Fixed-Time algorithm achieved an average throughput of 165.83 vehicles per minute, compared with 190.22 vehicles per minute for the Priority-based algorithm. This corresponds to an improvement of 14.7%.

The difference remained evident in the 30-minute simulations. The Fixed-Time algorithm processed an average of 164.44 vehicles per minute, whereas the Priority-based algorithm processed 190.78 vehicles per minute. This represents a 16.0% improvement.

Overall, the throughput results show that the Priority-based algorithm consistently processed more vehicles within the same period. The improvement increased from 12.4% in the 5-minute simulations to 16.0% in the 30-minute simulations. The higher throughput can be attributed to the algorithm's ability to allocate green time according to the traffic conditions observed at the approaches, reducing unnecessary green time for approaches with fewer waiting vehicles [14].

4.4 Fairness

Fairness was considered to evaluate whether the reduction in waiting time was distributed relatively evenly among the different traffic approaches and vehicle types. A traffic-control strategy that improves overall throughput but consistently disadvantages a particular approach may not provide a satisfactory solution [7].

The Priority-based algorithm determines signal priority based on the traffic conditions observed at the approaches [7]. As a result, approaches with longer queues can receive priority when required, while approaches with relatively lower traffic demand do not continuously retain the green phase. This allows the available signal time to be distributed according to the current traffic condition rather than according to a fixed sequence alone.

The experimental results indicate that the Priority-based approach produced a more balanced distribution of waiting time across the approaches. The Fixed-Time algorithm, on the other hand, maintains the same signal timing pattern regardless of changes in the number of waiting vehicles. Consequently, an approach with relatively higher traffic demand may experience longer queues while another approach is receiving a predetermined green interval.

The random selection of vehicle types also ensured that the evaluation was not restricted to a single vehicle category. Since vehicle types were randomly generated during the simulations, the observed performance reflects the behavior of the algorithms under the mixed traffic conditions defined in the simulation model.

4.5 Overall Comparison

The results from the four performance metrics show a consistent advantage for the Priority-based Dynamic Traffic Signal Control algorithm over the Fixed-Time algorithm.

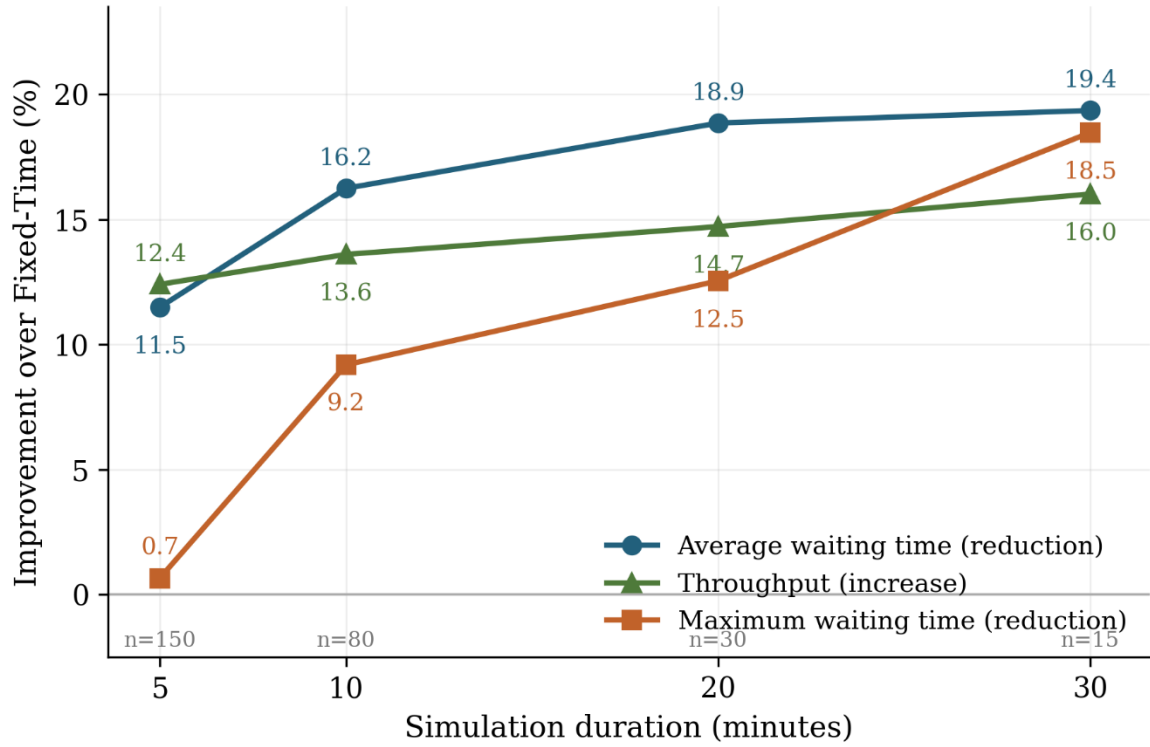


Figure 4.4 Overall Improvements of All Matrices.

As shown in Figure 4.4, the Priority-based algorithm improved intersection throughput by approximately 12–16% across the tested simulation durations. At the same time, the average vehicle waiting time was reduced from 11.49% in the 5-minute simulations to 19.35% in the 30-minute simulations.

A similar trend was observed for maximum waiting time. The reduction was approximately 0.7% for the 5-minute simulations and increased to 18.5% for the 30-minute simulations. Therefore, the improvement was not limited to the average behavior of vehicles; the Priority-based approach also reduced the worst-case waiting experienced by individual vehicles.

The results across the different metrics follow a similar pattern: the difference between the two algorithms becomes more pronounced as the simulation duration increases. This

suggests that the adaptive control mechanism becomes increasingly useful when traffic continues to accumulate. While the Fixed-Time algorithm continues to operate using predetermined signal durations, the Priority-based algorithm can respond to changes in queue conditions and allocate signal priority accordingly

4.6 Discussion

The experimental results show that the Priority-based Dynamic Traffic Signal Control algorithm performed better than the Fixed-Time algorithm under the traffic conditions used in this study. The proposed approach achieved lower average and maximum waiting times while also producing higher intersection throughput. The improvement was observed across all simulation durations, indicating that the benefit of adapting signal operation according to the current traffic condition was consistent rather than limited to a particular simulation period.

The reduction in waiting time can mainly be attributed to the queue-based priority mechanism of the proposed algorithm. Unlike the Fixed-Time algorithm, which provides each direction with a predetermined green duration regardless of the number of waiting vehicles, the Priority-based algorithm uses the available vehicle information to determine the service order and green duration. As a result, approaches with greater traffic demand can receive additional green time, reducing the accumulation of vehicles at those approaches. This behavior becomes more important as the simulation continues and the difference in queue sizes becomes more pronounced.

The throughput results support the same observation. The Priority-based algorithm processed more vehicles per minute than the Fixed-Time algorithm for all four simulation durations. The improvement increased from 12.4% in the 5-minute simulations to 16.0% in

the 30-minute simulations. This indicates that the reduction in waiting time was accompanied by an increase in the number of vehicles processed through the intersection. The increasing advantage over longer simulations also suggests that the adaptive mechanism is more effective when traffic demand is sustained and queues have more opportunity to develop.

The maximum waiting-time results provide an additional indication of the effectiveness of the proposed approach. While the difference between the two algorithms was small during the shortest simulation, the gap increased considerably in longer simulations. The maximum recorded waiting time decreased from 831 seconds with the Fixed-Time algorithm to 677 seconds with the Priority-based algorithm in the 30-minute case. This suggests that the proposed approach can reduce not only the general level of delay but also the occurrence of excessively long individual waiting times.

Despite these improvements, the results should be considered within the scope of the simulation environment. The experiments were conducted on a single four-way intersection under controlled traffic-generation conditions, with vehicles generated at a constant rate of 5 vehicles per second and vehicle characteristics determined by the defined random generation mechanism. Real-world traffic includes additional factors such as driver behavior, lane-changing, road geometry, pedestrian movement, and interactions between multiple intersections. Further evaluation using more complex traffic scenarios and real-world traffic data would therefore be necessary to determine the performance of the proposed algorithm under practical conditions.

Chapter 5. Conclusion

This study developed a VANET-based dynamic traffic signal control system for adaptive intersection management. The proposed system uses communication between On-Board Units (OBUs), Roadside Units (RSUs), and Access Points (APs) to obtain vehicle information and dynamically determine signal priority and green-time allocation according to traffic demand.

The proposed Priority-based Dynamic Traffic Signal Control algorithm was evaluated against a conventional Fixed-Time traffic signal control algorithm under the same simulated traffic conditions. The experimental results showed that the Priority-based algorithm consistently reduced vehicle waiting time and increased intersection throughput. The average waiting time reduction increased from 11.49% in the shortest simulation condition to 19.35% in the longest condition. The maximum recorded waiting time also decreased from 831 seconds for the Fixed-Time algorithm to 677 seconds for the Priority-based algorithm in the 30-minute case. In addition, intersection throughput improved from 12.4% to 16.0% across the tested simulation durations.

These results indicate that adapting signal operation according to the observed traffic demand can provide better intersection performance than maintaining fixed signal durations under the traffic conditions considered in this study. The improvement was particularly noticeable during longer simulation periods, when differences in traffic demand and queue accumulation became more pronounced.

The proposed approach also demonstrates that adaptive traffic signal control does not necessarily require complex computational techniques. A relatively simple queue-based

priority mechanism can use available vehicle information to adjust signal operation and achieve measurable improvements in waiting time and throughput. This is consistent with previous research showing the potential of connected-vehicle and V2X data for traffic-responsive signal control [6], [14], [20].

However, the findings should be interpreted within the limitations of the simulation environment. The experiments were conducted at a single four-way intersection under controlled traffic-generation conditions. The simulation assumed a constant vehicle generation rate and simplified vehicle behaviour, communication, and road conditions. Factors such as driver behaviour, lane changing, detailed road geometry, communication delays, packet loss, pedestrians, and interactions between multiple intersections were not modelled. Therefore, the results demonstrate the potential of the proposed approach within the defined simulation environment rather than establishing its performance in real-world traffic conditions.

5.1 Future Work

Several extensions can be considered to further develop the proposed system.

First, the system could be extended from a single intersection to multiple interconnected intersections. Coordinating traffic signals across several intersections could allow traffic conditions at one intersection to influence signal decisions at neighbouring intersections, providing a basis for larger-scale traffic management.

Second, predictive techniques such as artificial intelligence and machine learning could be investigated to forecast traffic demand before congestion becomes severe. Such approaches could complement the current reactive priority-based mechanism by allowing signal timings to be adjusted based on predicted traffic conditions [6], [14].

Third, the system could be evaluated using real-world traffic datasets and, where possible, field experiments. Real traffic data would allow the effects of vehicle behaviour, traffic fluctuations, road characteristics, and communication conditions to be investigated more realistically.

Fourth, future implementations could investigate more advanced communication and computing technologies, including edge computing and newer vehicular communication technologies. These technologies could potentially reduce communication and processing delays and improve the reliability of distributed traffic-control systems.

Finally, additional functionality could be introduced for emergency vehicles and autonomous vehicles. Emergency vehicle prioritization could allow the system to temporarily modify normal signal operation when an emergency vehicle approaches an intersection. Security and fault-tolerance mechanisms could also be incorporated to improve the reliability and trustworthiness of the communication between vehicles and roadside infrastructure [8].

Overall, this study demonstrates that a VANET-assisted, priority-based dynamic traffic signal control approach can improve traffic performance at a simulated intersection compared with a fixed-time strategy. The reductions in waiting time and the increase in throughput provide evidence that using real-time vehicle information for adaptive signal control is a promising approach. The proposed system therefore provides a practical foundation for further investigation of connected-vehicle-based intelligent traffic management systems.

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